Project Overview: Fandango Movie Ratings — Bias Analysis

1. Motivation

Fandango is both a movie-ticket retailer and a source of movie ratings on its own website. That dual role creates an obvious conflict of interest: a rating platform that also profits from ticket sales has a financial incentive to show higher ratings, since better ratings can drive more purchases. In 2015, FiveThirtyEight (538) investigated this exact question and published their findings, along with the underlying data, in the article *Be Suspicious Of Online Movie Ratings, Especially Fandango's*.

This project reproduces that investigation independently, using the original 538 datasets, to check whether the same conclusion holds up.

2. Objective

Determine, using `pandas` and data visualization:

1. Whether Fandango's **displayed star rating (STARS)** is systematically higher than the **true rating (RATING)** derived from the underlying HTML/vote data.
2. Whether Fandango's ratings — once normalized onto a common 0–5 scale — are inflated relative to Rotten Tomatoes, Metacritic, and IMDB ratings for the same films.

3. Data Sources

Both datasets are from 538's public GitHub repository (https://github.com/fivethirtyeight/data), collected around August 24, 2015.

`fandango_scrape.csv`

Every film scraped from Fandango.com.

Column	Description
FILM	Movie title (format: `Title (Year)`)
STARS	Star rating displayed on the Fandango site
RATING	Actual average rating value pulled from page HTML
VOTES	Number of user reviews at time of scraping

`all_sites_scores.csv`

Aggregate ratings for films that had a Rotten Tomatoes score, RT user score, Metacritic score, Metacritic user score, an IMDB score, **and** at least 30 fan reviews on Fandango.

Column	Description
FILM	Movie title

RottenTomatoes	RT Tomatometer (critics) score
RottenTomatoes_User	RT audience score
Metacritic	Metacritic critic score
Metacritic_User	Metacritic user score
IMDB	IMDb user score
Metacritic_user_vote_count	Metacritic user vote count
IMDB_user_vote_count	IMDb user vote count

4. Analysis Workflow

Part 1 — Exploring Fandango's Own Data

- Loaded `fandango_scrape.csv`; inspected shape, dtypes, and summary statistics.
- Plotted **RATING vs. VOTES** as a scatterplot to check whether popularity relates to rating, and computed the correlation matrix.
- Parsed the release **YEAR** out of each `FILM` title string and counted films per year.
- Identified the **10 most-voted films** and the count of films with **zero votes** (i.e., effectively unreviewed); these were removed to create a `fan_reviewed` subset containing only films with at least one real vote.

Part 2 — Displayed Rating vs. True Rating

- Plotted overlapping KDE distributions of `STARS` (displayed) vs. `RATING` (true), clipped to 0–5, to visualize how the two diverge.
- Created a `STARS_DIFF` column (`STARS - RATING`, rounded) to directly quantify the size of the discrepancy per film.
- Used a count plot to see how often each size of discrepancy occurred, and identified the single film with close to a full-star (1.0) inflation.

Part 3 — Fandango vs. Other Rating Platforms

- Loaded `all_sites_scores.csv` and explored its structure.
- **Rotten Tomatoes:** computed the difference between RT critic and RT user scores (`RT_Diff`), plotted its distribution (including the absolute-value version), and pulled the top 5 films where users rated much higher than critics, and vice versa.
- **Metacritic:** scatterplotted critic score vs. user score to check agreement between the two.
- **IMDB:** scatterplotted Metacritic vote count vs. IMDB vote count to compare audience size across platforms, and identified the single films with the highest vote counts on each site (surfacing outliers where a film is very popular on one platform but not deeply rated on another).

Part 4 — Merging & Normalizing

- Performed an **inner merge** of the Fandango and all-sites tables on `FILM`, keeping only films present in both datasets.
- **Normalized every platform's score onto Fandango's 0–5 scale:**
    - `RT_Norm = RottenTomatoes / 20`
    - `RTU_Norm = RottenTomatoes_User / 20`
    - `Meta_Norm = Metacritic / 20`

- `MetaU_Norm = Metacritic_User / 20`
- `IMDB_Norm = IMDB / 2`

- Assembled a `norm_scores` DataFrame containing Fandango's original `STARS` and `RATING` alongside every normalized competitor score, enabling an apples-to-apples comparison.

Part 5 — Comparing Distributions Across Platforms

- Plotted **KDE distributions of all normalized scores together** to see, at a glance, whether Fandango's distribution sits noticeably to the right of (i.e., higher than) the other platforms.
- Directly compared **Rotten Tomatoes critic scores vs. Fandango's displayed STARS**, since RT critics showed the most uniform, "textbook" distribution and made for the clearest contrast.
- Built a histogram overlay of all normalized scores, and a clustermap grouping films by rating pattern across platforms.
- Isolated the **10 worst-rated films by Rotten Tomatoes critics** and examined how each of those same films was rated across every other platform, including Fandango.
- Plotted the distribution of ratings across all sites specifically for those 10 worst films.

5. Key Findings

1. **Displayed ratings run higher than true ratings.** The `STARS` distribution sits to the right of the `RATING` distribution — Fandango was rounding up, not just to the nearest star but in a way that skewed favorable. At least one film showed almost a full star of inflation.
2. **Fandango's normalized rating distribution is unusually narrow and high**, clustering around 3–4.5 stars, in contrast to Rotten Tomatoes critics, whose scores span the full range far more evenly.
3. **Even the worst-reviewed films (by RT critics) still scored high on Fandango.** When the 10 lowest-rated films by Rotten Tomatoes critics were checked against Fandango, the Fandango score remained disproportionately high compared to every other platform.
4. **The clearest single example: *Taken 3*.** It carried an average rating of just **1.86** across other platforms (RT, Metacritic, IMDB) but was displayed at **4.5 stars** on Fandango — one of the largest platform-vs-Fandango gaps in the dataset.

6. Conclusion

The independent re-analysis reaches the **same conclusion as the original 538 investigation**: in 2015, Fandango's displayed ratings were **not a neutral reflection of audience sentiment**. They were skewed upward relative both to Fandango's own true underlying rating and to ratings on Rotten Tomatoes, Metacritic, and IMDB — a pattern consistent with a business incentive to show higher ratings in order to sell more movie tickets.

7. Limitations & Notes

- The dataset is a **snapshot from August 2015** and does not reflect any changes Fandango may have made to its rating methodology afterward.

- The `all_sites_scores.csv` file only includes films that had **at least 30 Fandango reviews**, so the comparison excludes very low-traffic titles.
- Normalization from a 0–100 or 0–10 scale to a 0–5 scale is a simplification; it assumes each platform's scale is linear and directly comparable, which may not perfectly hold across differing rating methodologies (critic vs. audience, vote-weighted vs. simple average, etc.).

8. Tools & Libraries

Purpose	Library
Data manipulation	pandas, NumPy
Visualization	Matplotlib, Seaborn
Environment	Jupyter Notebook

9. Credits

Original data and investigation: FiveThirtyEight, from Walt Hickey's article *Be Suspicious Of Online Movie Ratings, Especially Fandango's* (2015).